

Engineering Academy
Syllabus & Course Expectations
2026-2027



Instructors/Course Titles

- **Mr. Corey Eckhart-** Applications of Physics in Engineering
- **Ms. Andie Shore** - English for Engineering
- **Ms. Breanna Snyder-** PLTW Engineering Design and Development

Google Classroom Code: ux27tg6h

** Learning for all three courses will be managed through Google classroom**

Course Description

Engineering Design and Development (EDD) is the capstone course in the PLTW high school engineering program. It is an engineering research course in which students work in teams to research, design and develop an original solution to a valid open-ended technical problem by applying the engineering design process. At the end of the course, student teams must submit a final written report as well as report and defend their final solution. In addition, students will complete smaller group projects during the first semester. In addition, this course is a look to the world of engineering while examining it through the lens of physics. In this course we will examine the tools and processes that engineers use in their work. We will work on design and building challenges which get students to go through the engineering process themselves while applying the laws of physics to help us understand how to design a working model and, later, to help us understand how our designs work. Lastly, this course will help you to hone the critical reading and thinking, research, and writing you will need to succeed in your career. This course is also preparing students to write as engineers in professional environments. You will learn to synthesize information, find answers, and present ideas to some of the different audiences you will encounter. Learning to write well will enable you to present yourself in an authentic and professional manner, as well as guide you through processing the work you complete with internal reflection. You will learn how to communicate your knowledge, plans, and ideas in a professional manner, but more importantly, you will learn to do so with intellect and advanced diction. By the end of the course, students should be confident in utilizing this knowledge in the construction and completion of a writing portfolio and self assessment.

Classroom Materials

Students enrolled in Santa Monica High School are not required to pay any fee, deposit, or other charge for participation in an educational activity offered by the school or the Santa Monica-Malibu Unified School District, except as authorized by law and consistent with the California constitution and Hartzell v. Connell (1984) 35 Cal.3d 899. Donations are sought and accepted for various activities and supplies, and are at times critical to the continued success of classes and activities, but donations are always voluntary, and no child will be denied the right to fully and meaningfully participate in any activity because he/she has not donated money or supplies. For more information, go to <http://www.smmusd.org>.

Donation Request

State law requires us to provide a public education free of charge. Subject to certain limited exceptions, the right to a free public education means we cannot require students or their families to purchase supplies. Santa Monica High School will provide all necessary supplies for your child to have a quality educational experience.

Due to limited funding for necessary materials/supplies we ask that parents/guardians make a donation to the PLTW Engineering Class to purchase materials for the upcoming capstone projects, and for potential transportation costs to WET Invention Challenge. We are requesting a donation of \$50 per student, but any amount is greatly appreciated. These funds will make the 4th year engineering class better for all students. You may send this donation by check made out to Santa Monica High School ASB and write “ PLTW Engineering” in the memo.

We thank you in advance for your willingness to help support the program.

Topics:

	Unit 1:	Unit 2:	Unit 3:	Unit 4:
Engineering	WET Invention Challenge	Research Project: Justification and Design	Research Project: Build, Test & Reflect	Ethics in Engineering
English	Independent Reading College Essays/Personal Statement Writing <i>To Engineer is Human</i>	<i>Project Hail Mary</i> by Andy Weir <i>I, Robot</i> by Asimov Research Writing	<i>Hidden Figures</i> <i>Flowers for Algernon</i> Literary Analysis Writing	<i>Oppenheimer</i> Independent Reading Technical Writing for Engineering Careers Ethics Unit
Physics	Topics include: Kinematics, Dynamics, Statics (CAD)	Topics include: Reverse Engineering, 3D Printing Prep	Topics include: Experimental design, Experimental analysis	Topics include: Scientific research, Scientific writing

Expectations of Students

Students are expected to follow all classroom procedures as well as the school wide policies for behavior:

- **Classroom Policy:**

- Students are expected to participate in class enthusiastically and safely, and complete all assigned work.
- Students are expected to follow the [laboratory safety rules](#) during all laboratory projects
 - Student attire must be school appropriate, and appropriate attire must be worn to complete laboratory assignments (closed-toe shoes for any work in the engineering lab)
- Students are expected to have cell phones or other technology distractions put away or turned off during class.
- Students are expected to abide by the rules outlined in the safety contract.
- Academic Integrity: All work must be original and completed by you. Use of AI to complete assignments is prohibited. If an assignment is completed using any AI program, you will receive a 0 with no opportunity to make it up, and your parent/guardian will receive an email. If you find something hard to do, ASK us.

- **Attendance:**

- Students are expected to attend all class periods and will be marked “absent” if not in attendance.
- If you are absent it is your responsibility to check google classroom and complete the work that you missed. You may get additional help for a missed class during flex time or office hours. The best way to get missed work is to download the work from Google Classroom.

- **Technology:**

- Responsible phone use in this classroom will consist of three rules:
 - Phones and airpods/headphones go in the phone pockets before the bell rings
 - Improper use of or disregard of rules for phones will result in confiscation and phones will be returned at the end of class
 - Further misconduct will result in following the school wide policy of sending phones to house offices and/or parent retrieval

- **Group Work:**

- Because this course is so heavily project based and group work focused, it is imperative that you are present to work with your group. If your attendance falters (whether by choice or circumstance), it is your responsibility to communicate with your teammate(s) to ensure you are contributing equitably. If this does not (or cannot) happen, we may have to intervene and implement independent study, have you work by yourself on your own project, or if egregious unexcused absences are noted, you may be removed from the course.

Grading Categories

- Assessments (80%)
 - Assessments include Research Document Components, Projects, Presentations, Essays, Tests/Quizzes, and Physics Labs
 - Reassessments will be given for: Research Document Components, Essays, and Physics Labs
 - Students will be given the opportunity to resubmit any assessment listed above, for up to 80% of its maximum score.
 - Students will have ONE week from the date the original assessment is returned to submit revisions.
- Assignments (20%)
 - Assignments include classwork, homework, and readings.
 - Assignments will be graded based on the following types of feedback and may include (but are not limited to): points/grades on Aeries, in-class review of overall trends/patterns, self-correction, peer review, group work, class discussion, self reflections, teacher written or verbal comments.
 - Grace Period for Assignments – Assignments turned in after the deadline will be accepted until at least the first time the concept is assessed.
 - *Classwork does not fall under the grace period, and will need to be turned in the day it is assigned (by the end of the class period.)*

Grading Scale & Distribution

A	90-100%
B	80-89%
C	70-79%
D	60-69%
F	50-59%

Schoolwide Classroom Policies

- **Attendance Policy** (<https://samohi.smmusd.org/about-us/policies/attendance>)
 - Students will be allowed one day per day missed in order to make up work.
 - Students who are absent on a test or quiz day will take that assessment the following day. If a student is absent for an extended period of time, it is the student's responsibility to make arrangements with the teacher on the day that they return.
 - Students will be required to make up missed labs during a designated time (usually flex time) as decided on by the student and teacher.
- **Academic Honesty Policy** (<https://samohi.smmusd.org/about-us/policies/academic-honesty>)
 - Samohi believes that personal integrity is basic to all solid achievement and that students will reach their full potential only by being honest with themselves and others. Samohi also believes that academic integrity is basic to the progress of the school community toward rich learning and the respect of outside communities. Samohi expects all students to respect the educational purposes underlying all school activities and also expects administration, faculty, and staff to provide an environment that encourages honesty. Students caught cheating will receive a zero on the assignment.
- **Electronics and Cell Phone Policy** (<https://samohi.smmusd.org/about-us/policies/electronics-and-cell-phone-policy>)
 - Santa Monica High School allows the use of electronic devices for non-academic means on campus ONLY BEFORE SCHOOL, BETWEEN PERIODS, DURING LUNCH, and AFTER SCHOOL.
- **Minimum F Policy**
 - All assignments, assessments and exams within each grading period will use 50% as a minimum F grade.

Santa Monica High School
Engineering Academy
2026-2027
Syllabus Contract

Student Name: _____

Last

First

I have read and understand the *Course Syllabus*, and I agree to abide by all classroom policies. _____ (Student Initials)

I understand that it is my responsibility to find out the work I've missed when I am absent, and I understand that I must make up any missed assessment (quiz or test) on the day that I return to school in my designated class period. _____ (Student Initials)

I have read the *Safety Rules and Procedures*, and I agree to participate safely in all laboratory experiments and to participate enthusiastically in all activities. _____ (Student Initials)

Student Signature

Date

Guardian Signature

Date

By signing this paper, you verify that you have read, understood, and agreed to the policies set forth on the Engineering Academy Syllabus. This includes your understanding of the expectations for students, classroom policies, and grading. You understand that you must follow all safety rules, class rules and policies.

Additional Information:

1. Parent/Guardian email address:

2. How can I help your student succeed this year? What supports have helped them in the past?

3. What additional information would you like me to know about your student?